Markscheme

Mathematics: analysis and approaches
Practice question bank

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163. (a)

Consider $z = 27i$.

Find the cube root of z with the smallest positive argument, in the form $a + bi$, correct to 3 significant figures.

$$z = 27(\cos 90^\circ + i \sin 90^\circ)$$ *(M1)*

cube roots have modulus 3 and arguments $\dfrac{90^\circ + 360^\circ k}{3}$ for $k = 0, 1, 2$ *(M1)* **A1**

smallest positive argument: $k = 0 \Rightarrow 30^\circ$ *(M1)*

$$3(\cos 30^\circ + i \sin 30^\circ) \approx 2.60 + 1.5i$$ **A1**

[5 marks]